

# Cultural data science

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Johannes

```
# Load and Process Yield Curve Data
files <- list.files("yield", full.names = TRUE, pattern = "\\..csv$")
yield_curve <- files %>%
  map_df(read_csv)
yield_curve$Date <- as.Date(yield_curve$Date, format = "%m/%d/%y")
yield_curve <- yield_curve %>%
  mutate(spread = `10 Yr` - `3 Mo`)

# Aggregate Yield Data by Month for Twitter analysis
yield_monthly <- yield_curve %>%
  mutate(month = floor_date(Date, "month")) %>%
  group_by(month) %>%
  summarise(spread = mean(spread, na.rm = TRUE)) %>%
  rename(date = month)

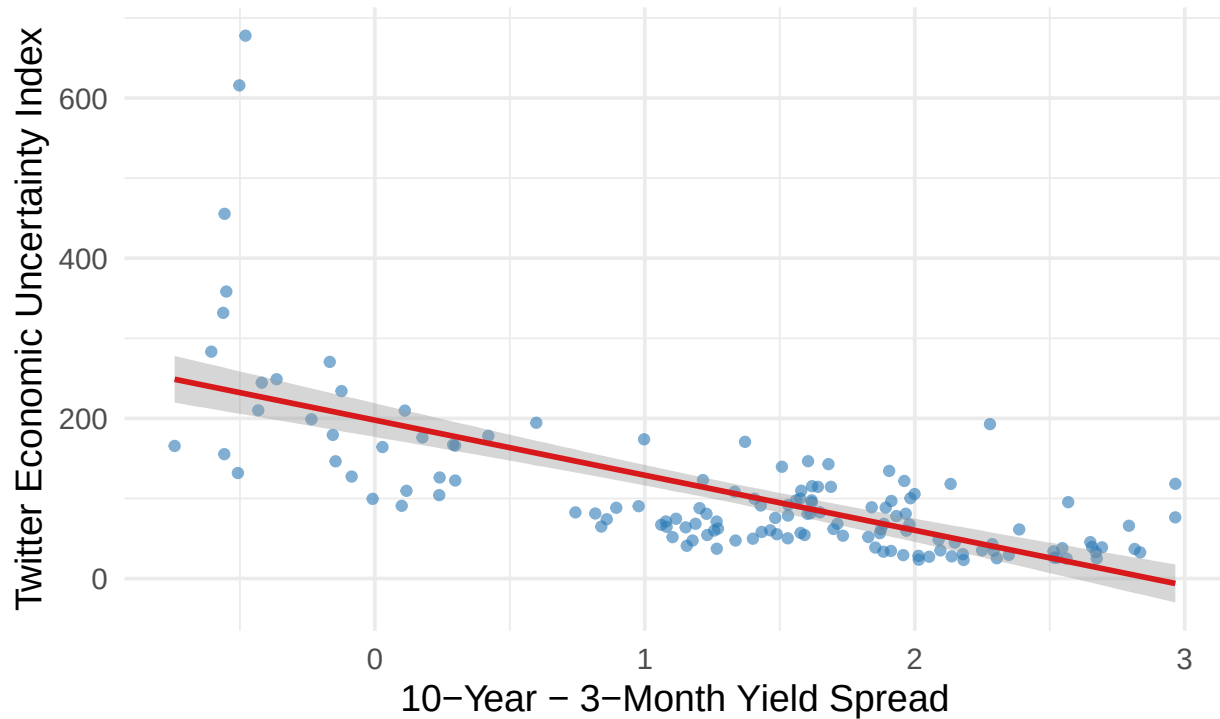
# Load Twitter Economic Uncertainty Data
twitter <- read_csv("Twitter_Economic_Uncertainty.csv")
twitter$date <- as.Date(twitter$date)
twitter_monthly <- twitter %>%
  mutate(month = floor_date(date, "month")) %>%
  group_by(month) %>%
  summarise(avg_teu = mean(`TEU-WGT`, na.rm = TRUE)) %>%
  rename(date = month)

# Merge Yield and Uncertainty Data
yield_twitter <- inner_join(twitter_monthly, yield_monthly, by = "date")

# Plot: Twitter Uncertainty vs Yield Spread
ggplot(yield_twitter, aes(x = spread, y = avg_teu)) +
  geom_point(alpha = 0.6, color = "#2C7BB6") +
  geom_smooth(method = "lm", se = TRUE, color = "#D7191C", linewidth = 1) +
  labs(
    x = "10-Year - 3-Month Yield Spread",
    y = "Twitter Economic Uncertainty Index",
    title = "Bond Yield Spread and Economic Uncertainty (2011-2023)",
    subtitle = "Monthly data showing relationship between yield curve and uncertainty"
  ) +
  theme_minimal(base_size = 14)
```

## Bond Yield Spread and Economic Uncertainty (2011–2020)

Monthly data showing relationship between yield curve and uncertainty



```
twitter_model <- lm(avg_teu ~ spread, data = yield_twitter)
summary(twitter_model)
```

```
##
## Call:
## lm(formula = avg_teu ~ spread, data = yield_twitter)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -100.93  -39.78   -7.25   22.65  447.13
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   197.809    10.596   18.67  <2e-16 ***
## spread        -68.776     6.425  -10.70  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 71.43 on 137 degrees of freedom
## Multiple R-squared:  0.4555, Adjusted R-squared:  0.4515
## F-statistic: 114.6 on 1 and 137 DF, p-value: < 2.2e-16
```

## Halfdan

```
# Aggregate Yield Data by Year for music analysis
yield_yearly <- yield_curve %>%
  mutate(Year = year(Date)) %>%
  group_by(Year) %>%
  summarise(spread = mean(spread, na.rm = TRUE))

# Load Billboard Hot 100 Data
df_bb <- read.csv("Hot100.csv")

# Join Billboard data with yearly yield data
df_analysis <- inner_join(df_bb, yield_yearly) %>% filter(Year < 2020)

# Mixed model: Danceability predicted by spread with random effect for year
mixed_model <- lmer(Danceability ~ spread + (1|Year), data = df_analysis)
summary(mixed_model)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Danceability ~ spread + (1 | Year)
## Data: df_analysis
##
## REML criterion at convergence: -286.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.79802 -0.67997  0.05046  0.65307  2.44629
##
## Random effects:
## Groups   Name                Variance Std.Dev.
## Year     (Intercept)  0.00000   0.0000
## Residual                    0.01839   0.1356
## Number of obs: 259, groups: Year, 19
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   0.681001   0.015981 257.000000  42.613   <2e-16 ***
## spread        -0.019445   0.008873 257.000000  -2.191    0.0293 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## spread -0.850
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
# For plotting, calculate yearly averages
df_year <- df_analysis %>%
  group_by(Year) %>%
  summarise(
```

```

    avg_dance = mean(Danceability, na.rm = TRUE),
    spread = mean(spread, na.rm = TRUE)
  )

# Plot: Danceability vs Yield Spread
ggplot(df_year, aes(x = spread, y = avg_dance)) +
  geom_point(size = 3, alpha = 0.7, color = "#2C7BB6") +
  geom_smooth(method = "lm", linewidth = 1.2, color = "#D7191C", se = TRUE) +
  labs(
    title = "Danceability and Yield Curve Spread",
    subtitle = "Does economic stress correlate with music preferences?",
    x = "Yield Curve Spread (pp)",
    y = "Average Danceability (0-1 scale)"
  ) +
  theme_minimal(base_size = 14)

```

